

AquaBophelo — Team Sprint Schedule & Task Allocations (Phase 4)

Project: AquaBophelo Water Monitoring & Truck Tracking System
Client Context: Sol Plaatje Municipality, Kimberley, Northern Cape
Sprint Period: 22 September 2026 – 28 September 2026
Final Phase 4 Deadline: 1 October 2026
Team Allocation:
- Backend Developers: Sisekelo "Cuba" Masombuka (Lead) & Nosipho Mbatha
- Frontend Developers: Nomcebo Nkosi & Jabulile Shabalala

Team Responsibilities Summary

Developer	Primary Domain	Core Responsibilities	Git Branch Prefix
Sisekelo (Cuba)	Backend Lead	Database Schema & EF Core Migrations, Identity & JWT Auth, Dams & Readings APIs, Alert Engine, Azure Deployment	backend/
Nosipho	Backend	Fleet Management APIs (Trucks/Routes), Trip Lifecycle APIs, SignalR TruckHub , Notification Service, Admin Reports	backend/
Nomcebo	Frontend	HCI Usability Heuristics, Accessibility (WCAG AA), Design System Tokens, Shared Layouts, Resident Dashboard, Dam Level Gauges, Chart.js	frontend/
Jabulile	Frontend	Mapbox GL JS Live Tracking Map , Driver Mobile Broadcast Screen (oversized touch targets $\geq 48\text{px}$), Geolocation Hook, Admin Command Center	frontend/

Day-by-Day Master Schedule (22 – 28 September 2026)

1. Tuesday, 22 September 2026 — Database Schema, Core Models & Layout Base

Sisekelo (Cuba) — Backend

- ✓ Create EF Core entities in `backend/AquaBophelo/Models/` : `ApplicationUser.cs` , `Area.cs` , `Dam.cs` , `DamReading.cs` .
- ✓ Create `AppDbContext.cs` extending `IdentityDbContext` .
- ✓ Create LocalDB instance `AquaBopheloDbInstance` and generate initial EF migration `InitialCreate` .
- ✓ Create `DbSeeder.cs` : seed Identity Roles (`Admin` , `Driver` , `Resident`), Sol Plaatje Municipal Areas (`Galeshewe` , `Kimberley Central` , `Roodepan`), Dams (`Newton Reservoir` , `Riverton Water Works`), and Admin User (`admin@aquabophelo.gov.za`).

Nosipho — Backend

- ✓ Create Fleet & Notification EF Core entities in `backend/AquaBophelo/Models/` : `Truck.cs` , `TruckRoute.cs` , `RouteStop.cs` , `Trip.cs` , `TripStop.cs` , `TruckLocation.cs` , `Alert.cs` , `AlertSubscription.cs` , `NotificationLog.cs` .
- ✓ Review foreign key constraints in `AppDbContext.cs` to prevent circular cascade deletes on `Truck` and `Trip` .

Nomcebo — Frontend (HCI Lead)

- ✓ Create shared UI components: `Sidebar.jsx` , `StatusBadge.jsx` (HCI accessibility rule: color + icon + text label), `StatCard.jsx` .
- ✓ Configure React Router (`App.jsx`) structure with responsive drawer navigation.

Jabulile — Frontend (Mapbox & Interactive Map)

- ✓ Scaffold interactive map component `LiveMap.jsx` using **Mapbox GL JS** with dark vector map style centered on Kimberley (`-28.7419` , `24.7719`).
- ✓ Build shared `AdminLayout.jsx` container.

2. Wednesday, 23 September 2026 — Authentication, JWT Tokens & Core REST APIs

Sisekelo (Cuba) — Backend

- ✓ Create `Dtos/AuthDtos.cs` (`RegisterDto` , `LoginDto` , `AuthResponseDto`).
- ✓ Create `Services/AuthService.cs` & `IAuthService.cs` (JWT token generation with claims for UserId, Email, Role).
- ✓ Create `Controllers/AuthController.cs` (`POST /api/v1/auth/register` , `POST /api/v1/auth/login` , `GET /api/v1/auth/me`).
- ✓ Create `Controllers/DamsController.cs` (`GET /api/v1/dams` , `GET /api/v1/dams/{id}` , `POST /api/v1/dams` , `PUT /api/v1/dams/{id}`).
- ✓ Create `Controllers/DamReadingsController.cs` (`GET /api/v1/dams/{id}/readings` , `POST /api/v1/dams/{id}/readings`).

Nosipho — Backend

- ✓ Create `Dtos/FleetDtos.cs` (`TruckDto` , `CreateTruckDto` , `RouteDto` , `CreateRouteDto` , `RouteStopDto`).
- ✓ Create `Controllers/TrucksController.cs` (`GET /api/v1/trucks` , `POST /api/v1/trucks` , `PUT /api/v1/trucks/{id}` , `PUT /api/v1/trucks/{id}/driver`).
- ✓ Create `Controllers/RoutesController.cs` (`GET /api/v1/routes` , `POST /api/v1/routes` , `PUT /api/v1/routes/{id}`).
- ✓ Create `Controllers/AreasController.cs` (`GET /api/v1/areas`).

Nomcebo — Frontend

- ☐ Create `src/auth/AuthContext.jsx` (token storage in `localStorage` , auto-attach Bearer token in `client.js`).
- ☐ Create `src/auth/ProtectedRoute.jsx` (route guards for Resident, Driver, and Admin roles).
- ☐ Create `src/pages/Login.jsx` & `src/pages/Register.jsx` (Resident self-registration with inline validation and HCI error prevention).
- ☐ Create `src/components/DamPanel.jsx` & `src/components/TrendChart.jsx` (Chart.js 7/30/90-day toggles).

Jabulile — Frontend

- ☐ Create `src/pages/admin/ManageDams.jsx` (Admin panel to manage dams and record manual water levels).
- ☐ Create `src/pages/admin/ManageTrucks.jsx` (Admin panel to manage water tankers and assign drivers).
- ☐ Create `src/pages/admin/ManageRoutes.jsx` (Admin panel to create routes and stops with coordinates).

3. Thursday, 24 September 2026 — Trip Lifecycle, Real-Time SignalR & Alert Engine

Sisekelo (Cuba) — Backend

- ✓ Create `Services/AlertService.cs` & `Controllers/AlertsController.cs` (`GET /api/v1/alerts` , `POST /api/v1/alerts`).
- ✓ Create `Controllers/SubscriptionsController.cs` (`GET` , `POST` , `DELETE /api/v1/subscriptions`).
- ✓ Create `BackgroundServices/AlertEvaluator.cs` : evaluates dam level changes on new readings and auto-generates Amber/Red alert records when thresholds cross (<50%, <30%, <15%).

Nosipho — Backend

- ☒ Create `Controllers/TripsController.cs` (`POST /api/v1/trips` , `GET /api/v1/trips/mine` , `POST /trips/{id}/start` , `POST /trips/{id}/stops/{stopId}/complete` , `POST /trips/{id}/end`).
- ☒ Upgrade `Hubs/TruckHub.cs` (`/hubs/trucks`):

Client $\rightarrow$ Server: `SendLocation(tripId, lat, lng, speed, heading)`

Server $\rightarrow$ Clients: `LocationUpdated` , `TripStarted` , `TripEnded` , `StopCompleted` .

Nomcebo — Frontend

- ☐ Create `src/pages/resident/ResidentDashboard.jsx` : dam level gauges, status badges, trend charts, active alert feed.
- ☐ Create `src/components/AlertSubscriptionsModal.jsx` (Resident SMS/Email area alert subscription modal).

Jabulile — Frontend (HCI Touch Targets & Driver Screen)

- ☐ Create `src/pages/driver/DriverTripScreen.jsx` : Mobile UI with **oversized touch targets ($\geq 48\text{px}$)** for one-tap operation while driving ("Start Trip", "Mark Stop Complete", "End Trip").
- ☐ Create `src/hooks/useGeolocationBroadcast.js` : uses `navigator.geolocation.watchPosition` to stream driver GPS updates to Nosipho's `TruckHub` .

4. Friday, 25 September 2026 — Mapbox Real-time Tracking & Notifications

Sisekelo (Cuba) — Backend

- ☐ Implement `INotificationService` with `DryRunNotificationService` (logs to `NotificationLog` table) and `SendGridEmailService` .
- ☐ Connect `AlertEvaluator.cs` to trigger automatic SMS/Email dispatch when dam levels change status bands.

Nosipho — Backend

- ☐ Optimize `TruckLocation` persistence: store latest position on `Truck` entity (`LastLatitude` , `LastLongitude` , `LastSeenAt`) for fast lookup, write full history records every 10 seconds.
- ☐ Create `Controllers/ReportsController.cs` (`GET /api/v1/reports/summary`) for admin analytics.

Nomcebo — Frontend

- ☐ Integrate live moving truck markers on **Mapbox GL JS** `LiveMap.jsx` for resident view using custom `@microsoft/signalr` hook (`useSignalR.js`).
- ☐ Add HCI Visibility of System Status indicator ("Connected", "Reconnecting...", "Disconnected").

Jabulile — Frontend

- ☐ Create `src/pages/admin/AdminOverview.jsx` : Command Center showing all active trucks moving across Kimberley on Mapbox GL JS map, dam status markers, and driver state badges.

- ☐ Create `src/components/BroadcastAlertModal.jsx` for Admin manual SMS/Email announcements.

5. Saturday, 26 September 2026 — System Integration, End-to-End Testing & Bug Fixes

👥 Sisekelo (Cuba) & Nosipho (Backend Integration)

- ☐ Review and merge feature branches into `main`.
- ☐ Test all endpoints in Scalar (`https://localhost:7154/scalar/v1`).

👥 Nomcebo & Jabulile (Frontend Integration)

- ☐ Merge feature branches into `main`.
- ☐ Verify complete navigation flow for all 3 user roles (Resident, Driver, Admin).

💛 Whole Team Collaboration

- ☐ Joint debugging session to fix Mapbox vector marker rendering, empty state handling, and error toasts.

6. Sunday, 27 September 2026 — Cloud Deployment (Azure + Vercel / Netlify)

👥 Sisekelo (Cuba) & Nosipho (Backend Deployment)

- ☐ Create **Azure App Service** instance for the ASP.NET Core API.
- ☐ Create **Azure SQL Database** (free student tier) and apply EF Core migrations (`dotnet ef database update`).

👥 Nomcebo & Jabulile (Frontend Deployment)

- ☐ Deploy `frontend/` directory to **Vercel** or **Netlify**.
- ☐ Configure environment variable `VITE_API_BASE_URL` pointing to deployed Azure API URL.

7. Monday, 28 September 2026 — Demo Telemetry, Simulator & Presentation Rehearsal

👤 Sisekelo (Cuba)

- ☐ Add `BackgroundServices/DamReadingSimulator.cs` for presentation demo.

Nosipho

- ☐ Export final Postman Collection and verify live Scalar documentation endpoints.

Nomcebo & Jabulile

- ☐ Final UI visual polish & HCI usability review (Nielsen Heuristics check).

Whole Team (Phase 4 Presentation Dry-Run)

- ☐ Conduct 2 full practice runs of the Phase 4 Capstone Presentation.